

A Linear Bijection Between 312-Avoiding Permutations and Dyck Paths

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1 Abstract

We present an explicit bijection between 312-avoiding permutations and Dyck paths. This map is defined via record increments. It provides a direct description of the correspondence between these two Catalan families and preserves their Tamari lattice structures. It carries several natural combinatorial statistics, such as records and peaks, and yields a combinatorial proof of their Narayana distribution. Its recursive description allows for an explicit inverse map. This bijection suggests further directions for studying other statistics and for extending the approach to other pattern-avoiding permutations.

2 Introduction

312-avoiding permutations are widely studied in combinatorics. They appear whenever tasks must respect precedence constraints: for example, a robot scheduling its actions, an RNA molecule folding, or a program evaluating its expressions. In each case, the 312-condition guarantees that no dependency is violated.

Dyck paths form a classical Catalan family, closely linked to binary trees and parenthesized expressions: structures that play a central role in combinatorics. They describe paths that never drop below zero: a model for processes that build up and release. For example, a call stack in computer science, a growing and shrinking population, or the trace of base-pair openings.

Although these two objects are counted by Catalan numbers, most known bijections between them are indirect, passing through binary trees or parenthesizations, losing the natural order during the procedure.

Bijections between Catalan families are a classical topic in combinatorics. The Tamari lattice, introduced by Tamari and further studied by Huang, arises from the classical correspondence between binary trees and Dyck paths. The bijection of Björner and Wachs, published in 1997, between pattern-avoiding permutations and binary trees was shown to be a lattice isomorphism.

Since then, numerous authors have studied bijections between pattern-avoiding permutations and Dyck paths. Many of these constructions proceed indirectly, most of the time by using binary trees.

Our map ϕ is an explicit bijection from 312-avoiding permutations to Dyck paths. It preserves the Tamari order and thus induces a lattice isomorphism between the weak order on $Av_n(312)$ and the Tamari lattice on Dyck paths. This means that two configurations that are one step apart on one structure correspond exactly to two configurations one step apart on the other. As a result, statistics or algorithms developed for Dyck paths transfer immediately to 312-avoiding permutations, and conversely, in a straightforward manner.

To situate our contribution within this landscape, we summarize in the following table the main known bijections between 312-avoiding permutations and Dyck paths.

Author	domain	codomain	Key invariant	preserve the Tamari order	complexity
Krattenthaler [Krat01]	132	Dyck path	Left to Right minima positions	No	$O(n^2)$
Callan [Call04]	321	Dyck path	partial sum (A \ D code)	No	$O(n)$
BJS [Bousquet-Mélou and al.]	321	Dyck path	excesses (positions)	No	$O(n^2)$
our bijection	312	Dyck path	record increments (climb length)	Yes	$O(n)$

We experimentally verified that none of the classical bijections between Dyck paths and pattern-avoiding permutations (Krattenthaler, Callan, BJS) preserves the Tamari order: for each of them, a counterexample appears for $n = 6$ or $n = 7$

We begin by introducing the necessary definitions and illustrating them with examples. We then construct our bijection and describe its inverse. After establishing that this construction yields a lattice isomorphism, we study the natural statistics carried by the bijection. Finally, we conclude with perspectives and possible extensions.

3 Preliminaries

3.1 Dyck words and binary trees

Definition 1 (Dyck word). *A Dyck word is a word $w \in \{u, d\}^*$ satisfying:*

- (D1) $|u(w)| = |d(w)|$ (the number of u 's equals the number of d 's)
- (D2) for every prefix w' of w , one has $|u(w')| \geq |d(w')|$

We denote by $\mathcal{D}_n$ the set of Dyck words of length $2n$. Its cardinality is the n -th Catalan number C_n .

Definition 2 (Height). *For a word $w \in \{u, d\}^*$, its height is defined by:*

$$h(w) = |u(w)| - |d(w)|$$

A Dyck word is precisely a word w such that $h(w) = 0$ and $h(w') \geq 0$ for every prefix w' of w .

Lemma 1 (Canonical decomposition). *Let $w \in \mathcal{D}_n$ be a non-empty Dyck word, which necessarily begins with u . Then there exists a unique pair (A, B) of Dyck words (possibly empty) such that:*

$$w = u \cdot A \cdot d \cdot B$$

where uAd is the shortest prefix of w that returns to height 0. This decomposition is unique.

Proof. Since w is non-empty and must begin with u (indeed, $h(\emptyset) = 0$ and condition (D2) prohibits w from starting with d), we may write $w = u \cdot w'$.

Let k be the smallest index such that the prefix of w of length $k + 1$ returns to height 0. Such an index exists because $h(w) = 0$.

We then set $A = w'[1 \dots k - 1]$ and $B = w[k + 2 \dots 2n]$ so that A consists of the letters of w between the positions 2 and k .

By minimality of k , the prefix uAd is the shortest prefix of w that returns to height 0, and therefore the decomposition $w = uAdB$ is unique. □

Definition 3 (γ map). We define recursively the map $\gamma : \mathcal{D}_n \rightarrow \mathcal{T}_n$ by:

- $\gamma(\emptyset) = \text{empty tree (denote by } \emptyset \text{)}$.
- $\gamma(w) = \text{Node}(\gamma(A), \gamma(B))$ where $w = uAdB$ is the canonical decomposition of w .

Fact 1 (Huang-Tamari). The Tamari order on binary trees is defined by $A \leq B$ if A can be obtained from B by a sequence of right rotations.

The Tamari order on Dyck paths is defined by $A \leq B$ if A can be obtained from B by a sequence of local rotations shifting a rise to the right.

The map $\gamma : \mathcal{D}_n \rightarrow \mathcal{T}_n$ is a lattice isomorphism between the Tamari order on $\mathcal{D}_n$ and the Tamari order on binary trees $\mathcal{T}_n$.

Example 1. Let $w = uAdB$ and $A = ud, B = ud$. Then

$$\gamma(w) = \text{Node}(\gamma(A), \gamma(B))$$

With: $A = ud, B = ud$

We first compute $\gamma(ud)$:

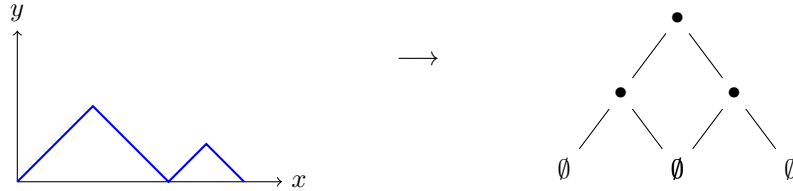
- the word $ud = u \cdot d$ has itself as the shortest prefix returning to height 0.
- hence its canonical decomposition is $ud = u \cdot \emptyset \cdot d \cdot \emptyset$.
- By definition, $\gamma(ud) = \text{Node}(\gamma(\emptyset), \gamma(\emptyset)) = \text{Node}(\emptyset, \emptyset)$, that is, a node whose two subtrees are empty.

We may now compute $\gamma(w)$:

Since $A = B = ud$, we obtain $\gamma(w) = \text{Node}(\gamma(A), \gamma(B)) = \text{Node}(\gamma(ud), \gamma(ud))$. So

$$\gamma(w) = \text{Node}(\text{Node}(\emptyset, \emptyset), \text{Node}(\emptyset, \emptyset))$$

This yields the following tree:



3.2 312-avoiding permutations

Definition 4 (Containing a pattern). A permutation σ of length n is said to contain a pattern μ of length $k \leq n$ if there exists a subsequence w of σ with $|w| = k$ such that the relative order of the values in w is the same as that of μ .

Definition 5 (312-avoiding permutation). A permutation σ of length n is 312-avoiding if it does not contain the pattern 312.

Equivalently, there do not exist indices $i_1 < i_2 < i_3$ such that:

$$\sigma_{i_2} < \sigma_{i_3} < \sigma_{i_1}$$

We denote by $Av_n(312)$ the set of permutations in S_n avoiding the pattern 312. We have $|Av_n(312)| = C_n$.

Definition 6 ($\beta_{\min}$ map). Let $\pi \in S_n$, ($n \geq 1$). Let $m = \min(\pi)$ and let k be its position in π . We write the unique decomposition $\pi = L \cdot m \cdot R$ where $L = (\pi_1, \dots, \pi_{k-1})$ and $R = (\pi_{k+1}, \dots, \pi_n)$. We define the map recursively by:

$$\beta_{\min}(\pi) = \text{Node}(\beta_{\min}(L), \beta_{\min}(R))$$

with $\beta_{\min}(\emptyset) =$ the empty tree.

Definition 7 (Right weak order). We define the right weak order $\leq_W$ in S_n by

$$\sigma \leq_W \pi \iff \text{Inv}(\sigma) \subseteq \text{Inv}(\pi)$$

where $\text{Inv}(\pi) = \{(i, j) : i < j, \pi_i > \pi_j\}$ is the set of inversions of π .

Fact 2 (Björner-Wachs). For every $\pi \in \text{Av}_n(312)$, the map $\beta_{\min}$ is a lattice isomorphism between $(\text{Av}_n(312), \leq)$ endowed with the right weak order and the poset of binary trees $\mathcal{T}_n$.

Definition 8 (Tamari order on $\text{Av}_n(312)$). We endow $\text{Av}_n(312)$ with the order induced by the right weak order $\leq_W$:

$$\pi \leq_W \sigma \iff \text{Inv}(\pi) \subseteq \text{Inv}(\sigma) \quad (\pi, \sigma \in \text{Av}_n(312))$$

Björner-Wachs showed that $\beta_{\min}$ transports the right weak order restricted to $\text{Av}_n(312)$ to the Tamari order on binary trees.

Example 2. Let $\pi = 213 \in S_3$

Step 1: Identify the minimum and decompose. The minimum of π is $m = 1$, appearing at position $k = 2$.

Thus $\pi = (2) \cdot 1 \cdot (3)$.

So, $L = (2), R = (3)$.

By definition: $\beta_{\min}(213) = \text{Node}(\beta_{\min}(2), \beta_{\min}(3))$.

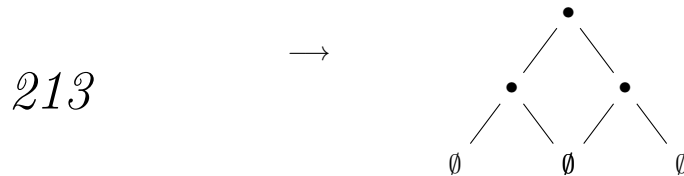
Step 2: Compute $\beta_{\min}(2)$. The permutation (2) has minimum at position 1.

Its decomposition is $\emptyset \cdot (2) \cdot \emptyset$.

Thus $\beta_{\min}(2) = \text{Node}(\beta_{\min}(\emptyset), \beta_{\min}(\emptyset))$, a leaf.

Step 3: Compute $\beta_{\min}(3)$. Similarly, $\beta_{\min}(3) = \text{Node}(\emptyset, \emptyset)$.

Step 4: Final assembly. $\beta_{\min}(213) = \text{Node}(\text{Node}(\emptyset, \emptyset), \text{Node}(\emptyset, \emptyset))$.



3.3 Record increase

While $\beta_{\min}$ provides a classical bijection from 312-avoiding permutations to binary trees, our goal is to construct a direct bijection to Dyck paths. To do so, we introduce the record-increase encoding.

Definition 9 (Record increase). Let $L = (l_1, \dots, l_p)$ be a finite sequence of positive, distinct integers not necessarily consecutive. For every $i \geq 1$, define:

$$M_i(L) = \max(l_1, \dots, l_i), \quad M_0(L) = 0$$

The record increase of L is the sequence $(\mu_1(L), \dots, \mu_p(L))$ defined by:

$$\mu_i(L) = \begin{cases} l_i - M_{i-1}(L) & \text{if } l_i > M_{i-1}(L) \\ 0 & \text{else.} \end{cases}$$

When $L = \pi \in S_n$, we simply write $\mu_i = \mu_i(\pi)$ and $M_i = M_i(\pi)$.

We then define:

$$\phi(L) = u^{\mu_1(L)} du^{\mu_2(L)} d \dots u^{\mu_p(L)} d$$

This definition applies to any sequence of positive distinct integers, and in particular to permutations.

Lemma 2 (Telescoping). For every permutation $\pi \in S_n$ and every $k \in \{1, \dots, n\}$:

$$M_k = \sum_{i=1}^k \mu_i$$

Proof. We proceed by induction on k

Base case: $k = 1$. π_1 is always a record because $M_0 = 0 < \pi_1$. Thus, $\mu_1 = \pi_1 - 0 = \pi_1 = M_1$.

Induction step: Assume $M_{k-1} = \sum_{i=1}^{k-1} \mu_i$. We distinguish two cases:

- If $\pi_k > M_{k-1}$: then $M_k = \pi_k$ and $\mu_k = \pi_k - M_{k-1}$. So:

$$\sum_{i=1}^k \mu_i = M_{k-1} + (\pi_k - M_{k-1}) = \pi_k = M_k$$

- If $\pi_k \leq M_{k-1}$: then $M_k = M_{k-1}$ and $\mu_k = 0$. Hence:

$$\sum_{i=1}^k \mu_i = M_{k-1} + 0 = M_{k-1} = M_k$$

□

Proposition 1 (Properties of μ). For every permutations $\pi \in S_n$, the sequence $(\mu_1, \dots, \mu_n)$ belongs to $\mathcal{R}_n$, ie:

- (C1) $\mu_i \geq 0$ for all i ;
- (C2) $\sum_{i=1}^n \mu_i = n$;
- (C3) for all $k \in \{1, \dots, n\}$, $\sum_{i=1}^k \mu_i \geq k$

Proof. (C1) By definition, $\mu_i = \pi_i - M_{i-1} > 0$ at record positions, and $\mu_i = 0$ otherwise.

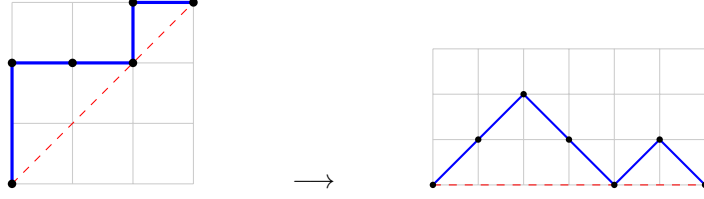
(C2) By the telescoping Lemma, $\sum_{i=1}^n \mu_i = M_n = \max(\pi_1, \dots, \pi_n) = n$. Since π is a permutation of $\{1, \dots, n\}$.

(C3) Again in by the telescoping lemma, $\sum_{i=1}^k \mu_i = M_k = \max(\pi_1, \dots, \pi_k)$. The k values $\pi_1, \dots, \pi_k$ are the k distinct integers from $\{1, \dots, n\}$, so their maximum is at least k .

Thus, $M_k \geq k$. □

Example 3. In the left diagram, the vertical steps represent the successive values of the sequence L . The dashed diagonal shows the current maximum M_i .

The right diagram shows the Dyck path obtained by applying ϕ , where each rise corresponds to a record increase.



Proposition 2. *For every permutation, $\pi \in S_n$, $\phi(\pi)$ is a Dyck word.*

Proof. The total number of u -steps is $\sum_{i=1}^n \mu_i = n$ by (C2), and the total number of d -steps is also n . Hence $h(\phi(\pi)) = 0$, which shows (D1);

For (D2), we check that the height remains positive after each block $u^{\mu_i}d$. After the first blocks k , the height is:

$$\underbrace{+1 + 1 \dots + 1}_{\mu_i \text{ times}} \underbrace{-1}_d$$

the height increases by μ_i and decreases by 1.

Let

$$\sum_{i=1}^k (\mu_i - 1) = \left(\sum_{i=1}^k \mu_i \right) - k \geq 0$$

by (C3) (because $\sum_{i=1}^k \mu_i \geq k$). Additionally, inside the block $u^{\mu_i}d$, the height increases by μ_i before it decreases by 1, so it stays positive all the time. Thus, $\phi(\pi)$ satisfies (D2). $\square$

4 Bijection ϕ

4.1 Recursive decomposition and standardization

Definition 10 (Standardization). *Let $L = (l_1, \dots, l_p)$ be a sequence of distinct integers. Standardization $\text{std}(L)$ is the unique permutation of $\{1, \dots, p\}$ such that:*

$$\text{std}(L)_i < \text{std}(L)_j \iff l_i < l_j$$

Standardization keeps the relative order of the elements, and in particular, pattern avoidance: if $\pi \in \text{Av}_n(312)$ and L is a subsequence of π , then $\text{std}(L) \in \text{Av}_{|L|}(312)$.

Lemma 3 (Values distribution). *Let $\pi \in \text{Av}_n(312)$. After renormalizing so that the minimum is 1. We write $\pi = L \cdot 1 \cdot R$ with $p = |L|$. So:*

$$\text{Val}(L) = \{2, \dots, p+1\}, \quad \text{Val}(R) = \{p+2, \dots, n\}$$

Proof. By construction, $\text{Val}(L)$ and $\text{Val}(R)$ are disjoint and $\text{Val}(L) \cup \text{Val}(R) = \{2, \dots, n\} \setminus \{1\}$. The values of π other than 1 are divided between L and R .

First point: $\text{Val}(L) \subseteq \{2, \dots, p+1\}$

We argue by contradiction that there exists $i_1 \in \{1, \dots, p\}$, such that $\pi_{i_1} \geq p+2$. Let $l = \pi_{i_1}$.

Because $l \geq p+2$, the interval $\{2, \dots, l-1\}$ includes at least p elements. But $\text{Val}(L) \setminus \{l\}$ is a set of $p-1$ distinct elements in $\{2, \dots, n\} \setminus \{l\}$, so $\text{Val}(L) \setminus \{l\}$ can only hold $p-1$ elements in $\{2, \dots, l-1\}$. There remains at least one element of $\{2, \dots, l-1\}$ that is not included in $\text{Val}(L)$, and therefore is included in $\text{Val}(R)$.

Let $r \in \text{Val}(R)$ with $1 < r < l$, and let $i_3 > p+1$, be the position of r in π . Let $i_2 = p+1$, the position of the value 1 in π . We have $i_1 < i_2 < i_3$ and

$$\pi_{i_2} = 1 < \pi_{i_3} = r < \pi_{i_1} = l$$

The subsequence $(\pi_{i_1}, \pi_{i_2}, \pi_{i_3})$ satisfies $\pi_{i_2} < \pi_{i_3} < \pi_{i_1}$, which constitutes a 312 pattern. This contradicts $\pi \in Av_n(312)$.

So $Val(L) \subseteq \{2, \dots, p+1\}$.

Second point: $Val(L) = \{2, \dots, p+1\}$

The set $\{2, \dots, p+1\}$ contains exactly p elements, and $Val(L)$ is a subset of $\{2, \dots, p+1\}$ of cardinality p . So $Val(L) = \{2, \dots, p+1\}$

Third point: By complementarity, $Val(R) = \{2, \dots, n\} \setminus \{2, \dots, p+1\} = \{p+2, \dots, n\}$

□

Corollary 1. *According to the hypothesis of Lemma 3, for all $i \in \{1, \dots, p\}$ and $j \in \{1, \dots, q\}$ with $q = |R|$:*

$$L_i = \text{std}(L)_i + 1, \quad R_j = \text{std}(R)_j + (p+1)$$

Proof. $Val(L) = \{2, \dots, p+1\}$ is a shift of $\{1, \dots, p\}$: standardization that subtracts 1 from each element. $Val(R) = \{p+2, \dots, n\}$ is a shift of $\{1, \dots, q\}$: the standardization subtracts $p+1$ from each element. □

Lemma 4 (Recursive decomposition of $\phi(\pi)$). *Let $\pi \in Av_n(312)$ with $\pi = L \cdot 1 \cdot R$ (after the normalization for the minimum to be 1), $p = |L|$, $q = |R|$. Thus:*

$$\phi(\pi) = u \cdot \phi(\text{std}(L)) \cdot d \cdot \phi(\text{std}(R))$$

Proof. Denote $(\alpha_1, \dots, \alpha_p)$ the record increases of the standardized permutation $\text{std}(L)$ and $(\beta_1, \dots, \beta_q)$ those of $\text{std}(R)$. We denote $m_i = M_i(\text{std}(L))$ and $b_j = M_j(\text{std}(R))$ the respective current maxima.

By definition of ϕ and by concatenation of $\pi = L \cdot 1 \cdot R$:

$$\phi(\pi) = \phi(L) \cdot u^{\mu_{p+1}(\pi)} d \cdot \phi(R)$$

We compute each of the three terms.

Central term: $\mu_{p+1}(\pi) = 0$

At the moment of reading the value 1 in position $p+1$, the current maximum is $M_p(\pi) = \max(Val(L)) = p+1$ according to Lemma 3. Because $1 < p+1$, the value 1 is not a record, so $\mu_{p+1}(\pi) = 0$ and the central block is $u^0 d = d$.

Computation of $\phi(L)$

By Corollary 1, $L_i = \text{std}(L)_i + 1$ for all i , so $M_i(L) = m_i + 1$ for all $i \geq 1$ and $M_0(L) = 0$.

We distinguish two cases according to i :

- $i = 1$: L_i is always a record because $M_0(L) = 0 < L_1$. We have:

$$\mu_1(L) = L_1 - 0 = \text{std}(L)_1 + 1 = \alpha_1 + 1$$

- $i \geq 2$: L_i is a record of L if and only if $L_i > M_{i-1}(L)$, this is equivalent to $\text{std}(L)_i + 1 > m_{i-1} + 1$, i.e. $\text{std}(L)_i > m_{i-1}$, i.e. $\text{std}(L)_i$ is a record of $\text{std}(L)$. Thus, for $i \geq 2$, L and $\text{std}(L)$ have exactly the same record positions. At record positions:

$$\mu_i(L) = L_i - M_{i-1}(L) = (\text{std}(L)_i + 1) - (m_{i-1} + 1) = \text{std}(L)_i$$

At non record positions: $\mu_i(L) = 0 = \alpha_i$

So we obtain:

$$\phi(L) = u^{\alpha_1+1} d \cdot u^{\alpha_2} d \cdots u^{\alpha_p} d = u \cdot (u^{\alpha_1} d \cdot u^{\alpha_2} d \cdots u^{\alpha_p} d) = u \cdot \phi(\text{std}(L))$$

Indeed, we extract the first u which corresponds to the shift of the first record.

Computation of $\phi(R)$

By Corollary 1, $R_j = \text{std}(R)_j + (p+1)$ for all j , so $M_j(R) = b_j + (p+1)$ for all $j \geq 1$, and $M_0(R) = 0$.

For all $j \geq 1$: R_j is a record of R if and only if $R_j > M_{j-1}(R)$, which is equivalent to $\text{std}(R)_j + (p+1) > b_{j-1} + (p+1)$, i.e. $\text{std}(R)_j > b_{j-1}$ i.e. $\text{std}(R)_j$ is a record of $\text{std}(R)$. At record positions:

$$\mu_j(R) = R_j - M_{j-1}(R) = (\text{std}(R)_j + (p+1)) - (b_{j-1} + (p+1)) = \text{std}(R)_j - b_{j-1}$$

At non record positions: $\mu_j(R) = 0 = \beta_j$

In contrast to the case of L , there is no shift in the first term because $M_0(R) = b_0$ and $(p+1)$ cancel out the difference. So:

$$\phi(R) = \phi(\text{std}(R))$$

Assembly

$$\phi(\pi) = \phi(L) \cdot d \cdot \phi(R) = u \cdot \phi(\text{std}(L)) \cdot d \cdot \phi(\text{std}(R))$$

□

4.2 Binary tree compatibility

Lemma 5. $\gamma(\phi(\pi)) = \text{Node}(\gamma(\phi(\text{std}(L))), \gamma(\phi(\text{std}(R))))$

Proof. Let $A = \phi(\text{std}(L))$ and $B = \phi(\text{std}(R))$. By Proposition 2, A and B are Dyck words (possibly empty if L or R is empty). By Lemma 4:

$$\phi(\pi) = u \cdot A \cdot d \cdot B$$

To apply the definition of γ , we must show that uAd is the smallest prefix of $\phi(\pi)$ that returns to height 0.

- After the initial u : $h(u) = 1$
- During the reading of A : For every prefix A' of A , because A is a Dyck word, we have $h(A') \geq 0$. The current height in uA is $1 + h(A') \geq 1 > 0$. So the height does not reach 0 while reading of A .
- After the final d of uAd : since A is a Dyck Word, its height is always non-negative and $h(A) = 0$. Thus, after reading uAd , the height is $1 + 0 - 1 = 0$

Therefore, uAd is the smallest prefix of $\phi(\pi)$ that returns to height 0. By definition of the canonical decomposition and of γ :

$$\gamma(\phi(\pi)) = \text{Node}(\gamma(A), \gamma(B)) = \text{Node}(\gamma(\phi(\text{std}(L))), \gamma(\phi(\text{std}(R))))$$

□

Lemma 6. For every sequence of distinct integers τ , we have $\beta_{\min}(\tau) = \beta_{\min}(\text{std}(\tau))$

Proof. We proceed by induction on $|\tau|$

Base case If τ is empty, both sides are the empty tree: $\emptyset$.

Induction step Let $m = \min(\tau)$ and write $\tau = L \cdot m \cdot R$. Then $\text{std}(\tau) = \text{std}(L) \cdot 1 \cdot \text{std}(R)$ because the minimum of τ becomes 1 in $\text{std}(\tau)$ and the relative orders of L and R are preserved by standardization.

By definition of $\beta_{\min}$

$$\beta_{\min}(\tau) = \text{Node}(\beta_{\min}(L), \beta_{\min}(R))$$

By the induction hypothesis:

$$\beta_{\min}(L) = \beta_{\min}(\text{std}(L)), \quad \beta_{\min}(R) = \beta_{\min}(\text{std}(R))$$

Thus $\beta_{\min}(\tau) = \beta_{\min}(\text{std}(\tau))$

□

4.3 Principal theorem

Theorem 1. *For every $n \geq 1$ and every permutation $\pi \in Av_n(312)$:*

$$\gamma(\phi(\pi)) = \beta_{\min}(\pi)$$

Proof. We proceed by induction on n .

Base case: $n = 1$, $\pi = [1]$, $L = \emptyset$, $R = \emptyset$. We have $\phi([1]) = u^1d = ud$. The canonical decomposition of ud is $u \cdot (\emptyset) \cdot d \cdot (\emptyset)$, so $\gamma(ud) = \text{Node}(\gamma(\emptyset), \gamma(\emptyset)) = \text{Node}(\emptyset, \emptyset) = \text{leaf}$ (a node whose two subtrees are empty).

Moreover, $\beta_{\min}([1]) = \text{Node}(\beta_{\min}(\emptyset), \beta_{\min}(\emptyset)) = \text{leaf}$.

The equality holds.

Induction step: Suppose that the theorem holds for all 312-avoiding permutations of size less than n . Let $\pi \in Av_n(312)$. After normalization, we write $\pi = L \cdot 1 \cdot R$.

Justification that $\text{std}(L), \text{std}(R) \in Av(312)$: Every subsequence of π is still 312-avoiding (any pattern in L or R would also be a pattern in π). Standardization preserves relative order, so it preserves pattern avoidance. Thus $\text{std}(L), \text{std}(R) \in Av(312)$.

By Lemma 5:

$$\gamma(\phi(\pi)) = \text{Node}(\gamma(\phi(\text{std}(L))), \gamma(\phi(\text{std}(R))))$$

Because $|\text{std}(L)| = |L| < n$ and $|\text{std}(R)| = |R| < n$, the induction hypothesis can be applied:

$$\gamma(\phi(\text{std}(L))) = \beta_{\min}(\text{std}(L)), \quad \gamma(\phi(\text{std}(R))) = \beta_{\min}(\text{std}(R))$$

By Lemma 6, $\beta_{\min}(\text{std}(L)) = \beta_{\min}(L)$ and $\beta_{\min}(\text{std}(R)) = \beta_{\min}(R)$.

Finally:

$$\gamma(\phi(\pi)) = \text{Node}(\beta_{\min}(L), \beta_{\min}(R)) = \beta_{\min}(\pi)$$

□

5 Inverse Bijection Ψ

We now show that the map ϕ allows an explicit inverse description.

We recall that, by Proposition 1, every permutation $\pi \in S_n$ is sent to a sequence $(\mu_1, \dots, \mu_n) \in \mathcal{R}_n$.

Conversely, we will show how to directly reconstruct the permutation from such a sequence.

Definition 11 (Map Ψ). *Let $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$. For all $i \geq 1$,*

$$M_i = \sum_{k=1}^i \mu_k$$

so that M_i represents the current record that the reconstructed permutation must have.

We recursively construct a sequence

$$\pi = (\pi_1, \dots, \pi_n)$$

We initialise with $\pi_1 = M_1 = \mu_1$ and set $R = M_1$ where R is the current record.

Suppose now, that the values $\pi_1, \dots, \pi_i$ are already constructed.

- *If $\mu_{i+1} > 0$, set $\pi_{i+1} = R + \mu_{i+1} = M_{i+1}$, and update the current record $R = \pi_{i+1}$*
- *If $\mu_{i+1} = 0$, set $\pi_{i+1} = \max(\{1, \dots, R-1\} \setminus \{\pi_1, \dots, \pi_i\})$.*

The resulting sequence is denoted $\Psi(\mu)$.

Lemma 7. *For every $i \in \{1, \dots, n\}$, after constructing the first i values of $\Psi(\mu)$ the current record $R = M_i = \sum_{k=1}^i \mu_k$ and it is the highest value constructed so far.*

Proof. We proceed by induction on i

Base case ($i = 1$): $\pi_1 = \mu_1 = M_1$. The current record is $R = M_1$, which is the highest constructed value because it is the only one.

Induction step: Suppose the property is in effect up to step i .

- If $\mu_{i+1} > 0$, then by the definition of the algorithm, $\pi_{i+1} = R + \mu_{i+1}$. Since $\mu_{i+1} > 0$, $\pi_{i+1} > R$. Moreover,

$$R + \mu_{i+1} = M_i + \mu_{i+1} = M_{i+1}$$

The new current record is M_{i+1} , which is also the highest built value.

- If $\mu_{i+1} = 0$, then the algorithm chooses a value strictly less than R . The current record is not updated and remains equal to $M_i = M_{i+1}$. Since the new value is strict less than R , R remains the highest built value.

In both cases, the property holds. □

Lemma 8. *The map Ψ is well defined, i.e. when $\mu_{i+1} > 0$, the set $\{x < R | x \notin \{\pi_1, \dots, \pi_i\}\}$ is not empty.*

Proof. According to Lemma 7, the current record is $R = M_i$. Since $\mu \in \mathcal{R}_n$, $M_i = \sum_{k=1}^i \mu_k \geq i$.

The integers strictly less than R are exactly

$$1, 2, \dots, R - 1$$

a total of $R - 1$.

By Lemma 7 again, among the i values already constructed, only one is equal to R , the current record. The remaining values of $i - 1$ are strictly less than R .

Thus, at most $i - 1$ integers of $\{1, \dots, R - 1\}$ have already been used.

But, $R - 1 \geq i - 1$. so at least one integer strictly less than R is not yet used.

Therefore, the considered set is non-empty, which shows that the construction is well defined. □

Lemma 9 (Block structure). *Let $\pi = \Psi(\mu)$. Consider two successive records $R_j < R_{j+1}$ in the construction of π . The values built between these two records form a decreasing block.*

This block is built first with the values strictly between R_j and R_{j+1} , then possibly from the values strictly less than R_j .

Proof. Consider a block associated with the current record R_j . According to the definition of the algorithm, while no new record is created, we have:

$$\mu_{i+1} = 0$$

and therefore:

$$\pi_{i+1} = \max(\{1, \dots, R_j - 1\} \setminus \{\pi_1, \dots, \pi_i\})$$

Thus, at each step, the chosen value is the highest available value still strictly less than the current record.

Therefore, the values obtained in this block are strictly decreasing.

We now describe which values appear in this block. At the moment when the new record R_{j+1} is created, all values between the two records:

$$R_j < x < R_{j+1}$$

become available for the first time, because they were previously greater than the current record.

Since the algorithm always chooses the greatest available value, these new values are necessarily used before the values already available from the previous blocks. After all values in the interval $]R_j, R_{j+1}[$ are used, the algorithm continues with the remaining values, which are all less than R_j .

Therefore, each block is of the form:

$$\underbrace{a_k > a_{k-1} > \dots > a_1}_{R_j < a_1 < R_{j+1}} > \underbrace{b_l > \dots > b_1}_{b_i < R_j}$$

The block is decreasing and satisfies the announced property. $\square$

Proposition 3. *For every sequence $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$ If $\pi = \Psi(\mu)$, then for every t*

$$\mu_i(\pi) = \mu_i$$

In particular,

$$\psi(\Psi(\mu)) = u^{\mu_1} \cdot d \cdot u^{\mu_2} \cdot d \dots u^{\mu_n} \cdot d$$

Proof. Let $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$ and set $\pi = \Psi(\mu)$.

For every i , denote $M_i = \sum_{k=1}^i \mu_k$.

By Lemma 7, after constructing the first i values of π , the current record is exactly M_i .

Consider now an index i .

- If $\mu_i > 0$, then, according to the definition of Ψ we have $\pi_i = M_i = M_{i-1} + \mu_i$.

Since $M_i > M_{i-1}$, the value π_i is a new record of π . According to the definition of ϕ ,

$$\mu_i(\pi) = \pi_i - M_{i-1} = M_i - M_{i-1} = \mu_i$$

- If $\mu_i = 0$, then the algorithm chooses a value strictly less than the current record M_{i-1} . Thus, $\pi_i < M_{i-1}$ and π_i is not a record.

According to the definition of ϕ , $\mu_i(\pi) = 0 = \mu_i$.

In both cases, $\mu_i(\pi) = \mu_i$.

The record-increase sequences of π and of μ coincide term by term, which shows

$$\phi(\Psi(\mu)) = \phi(\mu)$$

$\square$

Corollary 2. *The map Ψ is the inverse of ϕ . In particular, $\Psi(\mu) \in Av_n(312)$ for every $\mu \in \mathcal{R}_n$*

Proof. By Theorem 1, the map

$$\phi : Av_n(312) \rightarrow \mathcal{R}_n$$

is a bijection.

Proposition 3 shows that $\phi \circ \Psi = Id_{\mathcal{R}_n}$.

By uniqueness of the inverse of a bijection, we have

$$\Psi = \phi^{-1}$$

In particular, $\Psi(\mathcal{R}_n) \subseteq Av_n(312)$. $\square$

6 Lattice isomorphism

Theorem 2. *For every $n \geq 1$, the map ϕ is a lattice isomorphism*

$$\phi : (Av_n(312), \leq_W) \rightarrow (\mathcal{D}_n, \leq_{Tamari})$$

Proof. **Step 1: ϕ is a bijection**

By Theorem 1, $\gamma \circ \phi = \beta_{min}$. Since γ and β_{min} are bijections (Facts 1 and 2), we deduce $\phi = \gamma^{-1} \circ \beta_{min}$, which is a bijection.

Step 2: ϕ preserves the order in both directions.

Let $\pi, \sigma \in Av_n(312)$. We have:

$$\pi \leq_{Tamari} \sigma$$

$$\iff Inv(\pi) \subseteq Inv(\sigma) \quad (\text{definition of } \leq_W \text{ on } Av_n(312))$$

$$\iff \beta_{min}(\pi) \leq_{Tamari} \beta_{min}(\sigma) \quad (\text{Fact 2: } \beta_{min} \text{ is a lattice isomorphism})$$

$$\iff \gamma(\phi(\pi)) \leq_{Tamari} \gamma(\phi(\sigma)) \quad (\text{Theorem 1: } \gamma \circ \phi = \beta_{min})$$

$$\iff \phi(\pi) \leq_{Tamari} \phi(\sigma) \quad (\text{Fact 1: } \gamma \text{ is a lattice isomorphism})$$

Step 3: ϕ is a lattice isomorphism

ϕ is a bijection between two finite lattices that preserves the order in both directions. In a finite lattice, such a bijection necessarily preserves infima and suprema: for every $\pi, \sigma \in Av_n(312)$,

$$\phi(\pi \wedge \sigma) = \phi(\pi) \wedge \phi(\sigma), \quad \phi(\pi \vee \sigma) = \phi(\pi) \vee \phi(\sigma)$$

Since $x \leq y \iff x \wedge y = x$ in any lattice, ϕ is a lattice isomorphism. $\square$

We have constructed an explicit bijection ϕ between the 312-avoiding permutations and Dyck paths. This bijection is in fact a lattice isomorphism, which shows that the right weak order restricted to $Av_n(312)$ coincides with the Tamari order on the Dyck words.

7 Transported statistics by ϕ

Proposition 4 (Records $\longleftrightarrow$ Peaks). *Let $\pi \in Av_n(312)$. Let $rec(\pi)$ the number of records of π , i.e. the number of indices i such that $\pi_i > M_{i-1}$. Let $peak(w)$ the number of peaks of a Dyck word w , i.e. the number of factors ud in w . Then*

$$rec(\pi) = peak(\phi(\pi))$$

Proof. By definition

$$\phi(\pi) = u^{\mu_1} d u^{\mu_2} d \dots u^{\mu_n} d$$

By the definition of record increase, $\mu_i > 0$ if and only if π_i is a record.

Consider an index i

If $\mu_i > 0$, the corresponding block is

$$u^{\mu_i} d$$

The last u of this block is immediately followed by the d , so exactly one ud factor appears, i.e. exactly one peak.

If $\mu_i = 0$, the block reduces to a single d and contains no ud factor.

Thus each i with $\mu_i > 0$ contributes exactly one peak and no other peaks appear.

Therefore

$$peak(\phi(\pi)) = |\{i : \mu_i > 0\}| = rec(\pi)$$

□

Proposition 5 (Narayana records distribution). *For every integer $n \geq 1$ and $1 \leq k \leq n$*

$$|\{\pi \in Av_n(312) : rec(\pi) = k\}| = N(n, k)$$

where

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}$$

is the number of Narayana (The sum of the Narayana number in the n -th line of the Narayana triangle is equal to the n -th Catalan number. We have $\sum_{k=1}^n N(n, k) = C_n$).

Proof. The map

$$\phi : Av_n(312) \rightarrow \mathcal{D}_n$$

is a bijection between the 312-avoiding permutations and Dyck paths of semilength n .

By the previous proposition,

$$rec(\pi) = peak(\phi(\pi))$$

Thus ϕ preserve exactly the number of records and the number of peaks.

But the Narayana number $N(n, k)$ counts the Dyck paths of semilength n with exactly k peaks.

Since ϕ is a bijection, the number of 312-avoiding permutations with exactly k records is also $N(n, k)$.

Hence

$$|\{\pi \in Av_n(312) : rec(\pi) = k\}| = N(n, k)$$

□

Proposition 6 (Height at block endpoints). *Let $\pi \in Av_n(312)$.*

After reading the first k blocks

$$u^{\mu_1} d \dots u^{\mu_k} d,$$

the height of the path $\phi(\pi)$ is

$$M_k - k.$$

Proof. Let $\pi \in Av_n(312)$ et $M_k = \max(\pi_1, \dots, \pi_k)$

By the telescoping Lemma,

$$M_k = \sum_{i=1}^k \mu_i$$

After reading the first k blocks

$$u^{\mu_1} d \dots u^{\mu_k} d$$

The height of the path is

$$\sum_{i=1}^k (\mu_i - 1) = \left(\sum_{i=1}^k \mu_i \right) - k = M_k - k$$

Thus the heights reached at the block endpoints are exactly $M_k - k$. The maximal height at a block endpoint is

$$\max_{1 \leq k \leq n} (M_k - k)$$

The vertical geometry of $\phi(\pi)$ is entirely determined by the record sequence.

□

Remark 1. The quantity $M_k - k$ measures the gap between the largest value seen so far and the number of positions already read .

When records appear early, M_k grows quickly and the path rise high. When record are delayed, $M_k - k$ stays small and the path remains close to the axis.

Thus, the vertical geometry of the path $\phi(\pi)$ directly reflects the growth rate of records of π . In particular, the current maxima sequence

$$(M_1, \dots, M_n)$$

completely determines the height of the path at bloc endpoints.

Proposition 7 (maximal height). Let $\pi \in Av_n(312)$. The maximal height reached by the Dyck path $\phi(\pi)$ is

$$H(\phi(\pi)) = \max_{1 \leq k \leq n} (M_k - k) + 1$$

Proof. Write

$$\phi(\pi) = u^{\mu_1} d u^{\mu_2} d \dots u^{\mu_n} d$$

After reading the first $k - 1$ blocks, the height of the path is

$$M_{k-1} - (k - 1)$$

. The $k - ieme$ block is of the form $u^{\mu_k} d$. During the μ_k climbs, the height increases at each letter u of one unit. Thus the maximal height in this block is

$$M_{k-1} - (k - 1) + \mu_k$$

But, by definition of record increase,

$$\mu_k = M_k - M_{k-1}$$

We have

$$M_{k-1} - (k - 1) + \mu_k = M_k - k + 1$$

Since to each block reaches its maximum height just before the final descent, the height of the path is the maximal of the different block height. Thus,

$$H(\phi(\pi)) = \max_{1 \leq k \leq n} (M_k - k) + 1$$

□

Definition 12 (Runoff area). Let

$$w = u^{\mu_1} d u^{\mu_2} d \dots u^{\mu_n} d$$

a Dyck word.

Its area (in this article) is defined as the sum of the heights at which the descents start, i.e.

$$Area(w) = \sum_{k=1}^n h_k$$

where h_k designe its height just before the $k - ieme$ letter d .

Proposition 8 (Runoff area). Let $\pi \in Av_n(312)$. Then

$$Area(\phi(\pi)) = \sum_{k=1}^n (M_k - k + 1)$$

Proof. Write

$$\phi(\pi) = u^{\mu_1} du^{\mu_2} d \cdots u^{\mu_n} d$$

By the proposition 7, after reading the first $k - 1$ blocks, the height of the path is

$$M_{k-1} - (k - 1)$$

The $k - ieme$ block is of the form

$$u^{\mu_k} d$$

Just before the final descent of the block, the height is

$$M_{k-1} - (k - 1) + \mu_k$$

Since,

$$\mu_k = M_k - M_{k-1}$$

this equals

$$M_{k-1} - (k - 1) + \mu_k = M_k - k + 1$$

Summing over all k gives:

$$Area(\phi(\pi)) = \sum_{k=1}^n (M_k - k + 1)$$

□

The previous results show that natural geometric statistics of Dyck paths translate directly into the current maxima sequence (M_k) .

The bijection ϕ encodes not only the recursive structure of the 312-avoiding permutations, but also several of their statistics in a simple way.

8 Perspectives and conclusion

The results presented in this article show that the map ϕ provides an explicit and linear bijection between 312-avoiding permutations and Dyck paths.

This construction links only by record increments, making the bijection simple and algorithmically efficient: the map runs in $O(n)$ time and is straightforward to implement.

Beyond its explicit nature, ϕ reveals a structural correspondence. We proved that this bijection is a lattice isomorphism between the right weak order restricted to $Av_n(312)$ and the Tamari order on Dyck paths.

The bijection also transports several natural statistics. Records correspond to peaks and geometric quantities such as height and runoff area translate directly into the current maxima sequence (M_k) . Thus, ϕ not only encodes the structure of 312-avoiding permutations, but also preserves the combinatorial.

These observations open up several perspectives. One may investigate additional statistics or extend the construction to other pattern-avoiding classes. The simplicity of ϕ suggests that the increments in the records may serve as unifying principle for describing bijections between Catalan families.

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9 Annexe

For experimental tests, we have try in pthon/sage several lemmas, propositions and theoremes in this document. Here the link to the code in github and the outputs.

tests of permutations to $n = 11$

1. Test of corollary: check that $L = 2, 3, \dots, -L + 1$

No conter-example find (tests: 82499 permutations)

2. Test of the mu structure: check the recursivity of mu

always True

3. Test the indermediate conjecture: check the decomposition of phi with the minimum

succes: 82499/82499 tests pass

4. Test of lemme 2: check that $\Phi(\pi) = u + \Phi(\text{std}(L)) + d + \Phi(\text{std}(R))$

succes: 82499/82499 tests pass

5. Test the formula WL: check the direct construction of the left part of phi

succes: 82499/82499 tests pass

6. Test the formula L: check that $WL = u + \Phi(\text{std}(L))$

succes: 82499/82499 tests pass

7. Test of the principal theorem: $\gamma(\Phi(\pi)) = \beta_{min}(\pi)$

always True
